

The 14th International Conference on Emerging Ubiquitous Systems and Pervasive Networks
(EUSPN 2023)
November 7-9, 2023, Almaty, Kazakhstan

Advancements in Artificial Intelligence and Machine Learning for Stock Market Prediction: A Comprehensive Analysis of Techniques and Case Studies

Rihab Najem, Meryem Fakhouri Amr, Ayoub Bahnasse*, Mohamed Talea

LTI Laboratory, Faculty of Sciences Ben M'sick (FSBM), Hassan II University of Casablanca, Casablanca, Morocco

Abstract

Stock market forecasting is a classic but challenging problem that has attracted the attention of economists and computer scientists. The activity of trading involves high risks, the investors may lose a part of the totality of the amount they invested. Hence a need for more intelligent techniques to help make investment decisions. The purpose of this study is to provide first an overview of artificial intelligence and machine learning techniques used in recent studies for forecasting the stock market, then to present not only the different data types, commonly used evaluation metrics, and different neural network structures but also to provide a new proposition research method. Our objective is to help researchers stay abreast of the latest advances and help them easily replicate previous studies as a baseline.

© 2024 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the Conference Program Chairs

Keywords: Stock market; Artificial Intelligence; AI; forecasting; Machine Learning; ML; prediction; trading.

1. Introduction

Buying and selling financial assets or cryptocurrencies in various markets to make money by financing assets or so-called "trading" involves high risks, such as the risk of losing the amount invested, and may not be suitable for all traders. Cryptocurrency prices can be affected by external factors such as financial, political, or regulatory events, and they are extremely unstable. Before deciding to invest in financial instruments or cryptocurrencies, one must

* Corresponding author.

E-mail address: a.bahnasse@gmail.com

understand the risks and costs involved in trading financial markets, carefully consider investment objectives, trading experience, and risk tolerance, and look for professional advice if necessary. Predicting the price changes of cryptocurrencies helps to predict and control our day-to-day buying activities and manage our spending.

This allows us to develop strategic plans and prepare for the unexpected. In addition to a global vision, it is also necessary to manage the budget to foresee the evolution, plan the projects, and avoid being tossed around by numerous constraints. In the era of the emergence of automation technologies, it has become possible to make forecasted balances in real-time and with a degree of relevance exceeding the average. From the above, management of all expenses becomes a real benefit for financial monitoring and determining future directions. Professional traders have usually used computers to assist them in the global financial markets, which have become increasingly complex due to their large structures. It is estimated that algorithm-based trading now accounts for almost most world transactions. While well-known tools of trading are designed to buy and sell financial instruments in the shortest possible time. AI-based models look for the best times to purchase and sell over the long term. Because of the abundance of data available, AI can be precious support for trading. Technology is constantly evolving and there are already algorithms in the financial markets that are more efficient than humans.

This paper is set as follows: first, we describe the common data types used in stock market forecasting in Section 2, and later, we present the global Machine Learning (ML) techniques in Section 3. Then, we discuss the related studies of recent works in Section 4. After that, we describe the proposed research methodology adopted in Section 5. In the end, we conclude with a summary of the key points of this paper in Section 6.

2. Data Types for Stock Market Forecasting

Several types of data can be used to make stock market forecasts, market data or historical data are the most used types by researchers such as Indu Kumar et al. [1], W. Bao, J. Yue & Y. Rao [2], Mehar Vijh et al. [3], and Wen Fenghua et al. [4] refer to data covering all trading activities that take place on exchanges, such as open/high/low/close prices, volume, etc. (historical prices in lookback windows) and forecasts a target (such as the closing price for the next day). Textual data may also be used, which is textual data related to texts provided by individuals, such as news, and social media. As another type of data, these data are difficult to collect and process but can provide information not found in historical data that may be useful for prediction. Sentiment analysis such as Natural Language Processing (NLP) can be applied to this textual data and generate sentiment factors that can later be used for forecasting.

Macroeconomic data compiled by W. Bao, J. Yue & Y. Rao [2] are also a very interesting type because they reflect the economic conditions of a particular sector, region, or country such as the Gross Domestic Product (GDP), and Consumer Price Index (CPI), etc. These metrics are relevant to the stock market because they indicate the overall health of the stock market and can confirm or deny the quality of the stock market rally. Other authors such as Jiawei Long et al. [5], chose to use knowledge graph data because there is a strong relationship between different markets and different companies, for example, the stock movements of the same industry will be affected by the news. Open-source knowledge graph data is now available to improve predictive performance, supported by recently developed graph neural networks.

3. Machine Learning Algorithms

3.1 Support Vector Machine (SVM)

Support Vector Machine (SVM) is an algorithm of ML that is used for regression and classification problems. It transforms the data using a kernel trick method and then finds the best bound between possible outputs based on that information. The SVM tries to find the line separating the two classes. The new point is then classified according to whether it is on the positive or negative side of the line, depending on the class to be predicted. Indian Kumar et al. [1] Predicted the stock market using SVM, and then Wen Fenghua et al. [4] decomposed with Singular Spectrum Analysis (SSA), which gave them better results.

3.2 Random Forest (RF)

Random Forest (RF) is a set of classification or regression trees, constructed using guided sampling of training data and randomly selected features in tree induction. Predictions are made by aggregating (majority or average) the predictions of the set. Based on the predictions of the decision tree, the random forest model determines the outcome. A higher number of trees generally means higher accuracy. The best way to speed up the random forest is to reduce the number of estimators. The best way to improve the accuracy of the model is to increase the number of trees. After training the model, each tree in the forest is given a classification of the remaining data, and this tree is said to vote for that class. RF can perform regression and classification tasks. RFs provide good predictions that are easy to understand. It can handle large amounts of data efficiently. Compared to the decision tree algorithm, the RF algorithm provides higher accuracy in predicting the outcome. Several studies have used RF for stock market forecasting, including Indu Kumar et al. [1], Mehar Viji et al. [3], M. Azmi, G. C Runger & A. Berrado [6], and Mahla Nikou et al. [7].

3.3 K-Nearest Neighbor (KNN)

The K-Nearest Neighbor (KNN) algorithm used by Indu Kumar et al. [1] is a data classification method used to estimate the probability of a data point being a member of one or the other group based on the group to which the nearest data point belongs. Artificial Neural Network (ANN) algorithms use feature similarity to predict the value of each new data point. This means that each new point gets a predicted value based on its similarity to points in the training set. ANNs are useful when labeled data is too expensive or unavailable, enabling high accuracy in a variety of prediction problems. ANN is a simple algorithm based on the local minima of the objective function to learn unknown functions with the desired precision and accuracy. ANN directly uses the training dataset to make predictions. Predict new instances by searching the training set for the K neighbors that are most similar and summarize the output variables of these K instances.

3.4 Naïve Bayes (NB)

Naïve Bayes (NB) is a classification algorithm that uses Bayes' theorem. It predicts the probability of belonging to each class. The most likely class is the class with the highest probability. NB works on categorical input variables rather than numerical variables. It is useful for forecasting and forecasting data based on historical results. For real-time prediction, we consider NB as a fast-learning classifier. Therefore, it can be used to make real-time predictions. Indu Kumar et al. [1] also conducted a stock market prediction test on NB.

3.5 Interpretable regularized class association rules algorithm for classification (RCAR)

RCAR [6] produces high accuracy and understandable classifiers in rules space. The rules are mined from categorical data spaces. RCAR algorithm creates classifiers based on Regularized Class Association Rules [8]. This algorithm proceeds in three steps: First, mining a collection of Class Association Rules using predetermined support and confidence thresholds. The second step is to use a regularized logistic regression technique with a Lasso penalty to develop a model that forecasts the conditional likelihood that an event will occur. Because Lasso regularization is selective, useless rules are pruned in this step [9][10]. Third, using metarules and Lasso regularization, organize and visualize the CARs that make it through the initial stage of pruning. RCAR shows high performance and accuracy in several fields, especially in profiling and preventing security attacks [11]–[13].

3.6 Long Short-Term Memory

LSTM stands for "Long Short-Term Memory". LSTM units are used to learn recurrent neural networks "RNNs" that predict the future from sequences of different lengths. Note that RNNs work with any kind of sequence data, it is not limited to time series. LSTMs use a series of gates to control how information enters, stores and exits the network as sequences of data. A typical LSTM has three gates: the input gate, which contains information about the cell state,

the forget gate, which deletes information that is no longer necessary for the model, and the LSTM's output gate, which selects the information to be presented as an output. These gates can be thought of as filters, each forming its neural network. This algorithm is widely used in time series forecasting as described by W. Bao, J. Yue & Y. Rao [2], M. Nikou, G. Mansourfar & J. Bagherzadeh [7], and Taewook Kim & Ha Young Kim [14].

3.7 Artificial Neural Network (ANN)

A neural network is a computer learning system that uses a functional network to understand one form of data input and convert it into the desired output, usually another. An Artificial Neural Network (ANN) is an experiment to simulate the network of neurons of the human brain so that computers can learn and make decisions similar to humans. Artificial neural networks work by programming ordinary machines to behave as if they are brain cells interconnected with each other, developed by Wen Fenghua et al. [4] Neural networks are best suited for predictive analytics due to hidden layers. The algorithm linear regression uses only input and output nodes to predict. Neural networks also use hidden layers to make more accurate predictions. It learns like a human.

4. Related Work

The financial sector holds a high regard for novel technologies resulting from the progress made in management information technology. Many axes have been discussed by Rihab Najem et al. [15]. Trading is a hot topic in AI improvement today. To this end, several studies have been carried out in this direction.

On the one hand, Indu Kumar et al. [1] Tested several ML models such as SVM, RF, KNN, and NB using historical data. The architecture of the proposed implementation work consists of four steps, including the extraction of features from the dataset, classification of the training and test dataset, then finally the result evaluation. The forecast uses twelve technical indicators: moving averages: (MA10, MA50), which are responsible for smoothing stock price signals and facilitating trend identification, and rates of change (RoC1, RoC2), which simply measure the variation of price rate from time to time. Relative Strength Index (RSI), which detects if a stock is oversold or overbought, Volatility of 10, which is a measure of the return spread of a given stock. Williams%R, a Momentum indicator, reveals the position of the closing price relative to the peak, the Stochastic Oscillator, describes the position of the closing price compared to the low and high area, Channel Commodity Index (CCI) calculates the level of the actual price relative to the average price over time, Disparity (Disparity5, Disparity10), which measures the relative position of the selected moving average relative to the last closing price, and a price-volume trend that links volume to price.

The results show that the Random Forest algorithm performs better than all other algorithms according to the values of accuracy when the dataset is larger, while the Naive Bayes algorithm is less accurate when the size of the dataset is almost reduced by half. Likewise, the accuracy of any algorithm in predicting stock market trends may be reduced by reducing the number of technical indicators. The study by Wei Bao et al. [2] proposed a new deep learning solution that combines LSTM, Wavelet Transform (WT), and Stacked AutoEncoders (SAEs) for stock price forecasting. First, the WT decomposes the time series of stock prices to remove noise. Second, SAE is used to generate advanced deep features that predict stock prices. Third, an advanced denoising function is fed into the LSTM to predict the closing price for the next day. Prediction accuracy was assessed using three measures, Theil's inequality coefficient (Theil U), correlation coefficient (R), and Mean Absolute Percentage Error (MAPE).

This study is the first work to initiate SAE methods to extract depth-invariant daily features from financial time series which is the main contribution to the community. RF and ANN techniques were proposed by Mehar Vijh et al. [3] and used to predict the next-day closing prices for several companies from different industries. New variables were created using financial data, OHLC (Open, High, Low, and Close) prices, that are used as input to the model. The historical data set available on the company's website contains only a few features, such as OHLC values of stock prices. So, they use new variables to achieve greater accuracy in predicting prices. To predict the next day's closing prices, they used ANN, and RF is also used for comparative analysis. Predicted closing prices are subject to Root Mean Squared Error (RMSE), MAPE, and Mean Bias Error (MBE) to find the final minimum error in the predicted price. A comparative analysis based on MBE, MAPE, and RMSE values proves that ANN provides better stock price predictions compared to RF. In their work, Wen Fenghua et al. [4] decomposed the original stock index in terms of

trend, market volatility, noise, and time series by using SSA decomposition, the main idea is to get a set of singular values that contain information.

Further, the original sequence is passed through Singular Value Decomposition (SVD) and then picks different singular values to construct a series of different components. SVM predictions are made to obtain predicted values for these terms; they then use, as an input of the model, the predicted value for each term, and the actual price at that point as output to create a learning model. And use the learned model for predictions; where the predicted values for each component are the input variables, and the final predicted value is the output. Finally, they compared the forecasting effect of the combined SSA-SVM forecast and the SVM forecast and found that decomposing the original indicators into combined forecasts that have some economic impact on the SVM is more effective than the SVM forecast; and the SSA can better capture the characteristics of the series of the original index. The table below summarizes the methods used by the authors above and the best methods results given by each one of them.

Table 1. Comparative table of used techniques.

Authors	Best Accuracy	ML Algorithms								
		SVM	RF	KNN	NB	LSTM-WT-SAE	ANN	SSA-SVM	LSTM	LSTM-WT
Kumar et al. [1]	-RF for largest datasets. -NB for Small datasets.	X	X	X	X					
W. Bao, J. Yue & Y. Rao [2]	LSTM-WT-SAEs					X			X	X
Mehar Vijih et al. [3]	ANN		X				X			
Wen Fenghua et al. [4]	SSA-SVM	X						X		

5. Proposed Research Methodology

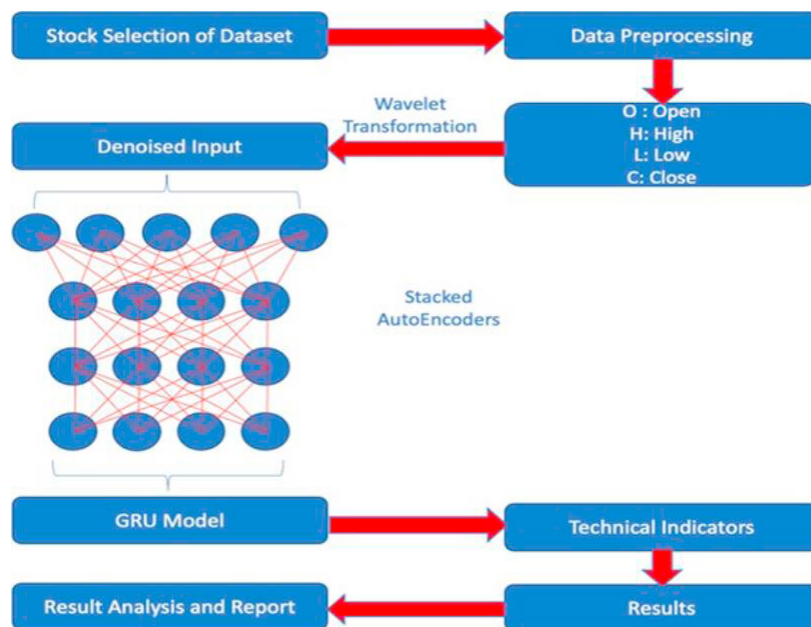


Fig. 1. Stock Market Forecasting Methodology Proposition.

The suggested stock market forecasting method and its description are shown in Fig. 1. First, we collect historical data from Yahoo Finance, the most common data source in finance, where we can find the OHLC prices. We preprocess these data and then apply WT to denoise the data, as it can handle non-stationary financial time series data. The main characteristic of WT is that compared with the Fourier Transform, it can simultaneously analyze the financial time series' frequency components. Therefore, wavelets are useful for dealing with very volatile financial time series.

The autoencoder is then applied to multi-layer training of technical indicators and OHLC variables while adopting SAE as the corresponding architecture of a deep neural network.

A single-layer SAE is considered a three-layer neural network; the first is the input layer, the second is an n-unit hidden layer designed to generate deep features for this single-layer neural network, and the third layer is the k-unit reconstruction layer. The goal of training a single-layer SAE is to reduce the error between the reconstructed vector and the input vector. Then we apply the Gated Recurrent Unit (GRU) model similar to LSTM, except the main difference between LSTM and GRU is that the GRU has two gates, one to reset and the other to update, while LSTM has three gates, input gate, output gate and forget gates. GRU is less complicated than LSTM because it has fewer gates. The computation performed by the GRU is faster and easier. However, the latter's capability/efficiency is no longer proven and is used the same as LSTM. We then proceeded to evaluate forecast accuracy using the five most popular measures RMSE, MAPE, MBE, Theil U, and R accuracy to analyze the results and generate a global report.

6. Conclusion

Artificial intelligence has become an increasingly important buzzword for business professionals and experts in recent years [16]. Taking an example of the stock market which is volatile; however, it presents a huge opportunity for investors to evaluate their funds. Artificial intelligence algorithms help investors do this using machine learning models. These models can quickly process different types of data, such as historical data, textual data, macroeconomic data, etc., and provide the right course of action for the future. Inspired by the fast development and the increasing use of machine learning algorithms for stock market forecasting, we review recent advances and describe a new combination of Wavelet Transform, Stacked AutoEncoders, and Gated Recurrent Unit models. Therefore, these findings can contribute as an initial guideline for future contributions.

References

- [1] Kumar, I., Dogra, K., Utreja, C., & Yadav, P. (2018, April). A comparative study of supervised machine learning algorithms for stock market trend prediction. In *2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)* (pp. 1003-1007). IEEE.
- [2] Bao, W., Yue, J., & Rao, Y. (2017). A deep learning framework for financial time series using stacked autoencoders and long-short term memory. *PloS one*, **12**(7), e0180944.
- [3] Vijh, M., Chandola, D., Tikkiwal, V. A., & Kumar, A. (2020). Stock closing price prediction using machine learning techniques. *Procedia computer science*, **167**, 599-606.
- [4] Fenghua, W. E. N., Jihong, X. I. A. O., Zhifang, H. E., & Xu, G. O. N.G. (2014). Stock price prediction based on SSA and SVM. *Procedia Computer Science*, **31**, 625-631.
- [5] Long, J., Chen, Z., He, W., Wu, T., & Ren, J. (2020). An integrated framework of deep learning and knowledge graph for prediction of stock price trend: An application in Chinese stock exchange market. *Applied Soft Computing*, **91**, 106205.
- [6] Azmi, M., Runger, G. C., & Berrado, A. (2019). Interpretable regularized class association rules algorithm for classification in a categorical data space. *Information Sciences*, **483**, 313–331. <https://doi.org/10.1016/j.ins.2019.01.047>
- [7] Nikou, M., Mansourfar, G., & Bagherzadeh, J. (2019). Stock price prediction using DEEP learning algorithm and its comparison with machine learning algorithms. *Intelligent Systems in Accounting, Finance and Management*, **26**(4), 164-174.
- [8] Azmi, M., & Berrado, A. (2015, October). Towards an interpretable rules ensemble algorithm for classification in a categorical data space. In *2015 10th International Conference on Intelligent Systems: Theories and Applications (SITA)* (pp. 1-6). IEEE.
- [9] Azmi, M., & Berrado, A. (2016, November). Class-association rules pruning using regularization. In *2016 IEEE/ACS 13th International Conference of Computer Systems and Applications (AICCSA)* (pp. 1-7). IEEE.
- [10] Azmi, M., & Berrado, A. (2021). CARs-RP: Lasso-based class association rules pruning. *International Journal of Business Intelligence and Data Mining*, **18**(2), 197-217.
- [11] Eddermoug, N., Mansour, A., Sadik, M., Sabir, E., & Azmi, M. (2021, June). KLM-based profiling and preventing security attacks for cloud computing: A comparative study. In *2021 28th International Conference on Telecommunications (ICT)* (pp. 1-6). IEEE.
- [12] Eddermoug, N., Sadik, M., Sabir, E., Mansour, A., & Azmi, M. (2019, June). Ppsa: Profiling and preventing security attacks in cloud computing. In *2019 15th International Wireless Communications & Mobile Computing Conference (IWCMC)* (pp. 415-421). IEEE.
- [13] Eddermoug, N., Mansour, A., Sadik, M., Sabir, E., & Azmi, M. (2019, October). Klm-PPSA: Klm-based profiling and preventing security attacks for cloud environments. In *2019 International Conference on Wireless Networks and Mobile Communications (WINCOM)* (pp. 1-7). IEEE.
- [14] Kim, T., & Kim, H. Y. (2019). Forecasting stock prices with a feature fusion LSTM-CNN model using different representations of the same data. *PloS one*, **14**(2), e0212320.

- [15] Najem, R., Amr, M. F., Bahnasse, A., & Talea, M. (2022). Artificial Intelligence for Digital Finance, Axes and Techniques. *Procedia Computer Science*, **203**, 633-638.
- [16] Azmi, M., Mansour, A., & Azmi, C. (Forthcoming in 2023). A Context-Aware Empowering Business with AI: Case of Chatbots in Business Intelligence Systems. *Procedia Computer Science*. [Preprint available: https://drive.google.com/file/d/1hWhJrWC-GjUvyib9krCPpFhymP27vwZ_/view?usp=sharing]